

Programme: B E – Information Science & Engineering
Internal Assessment – II

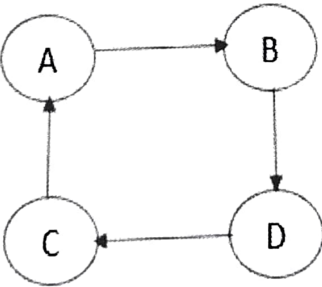
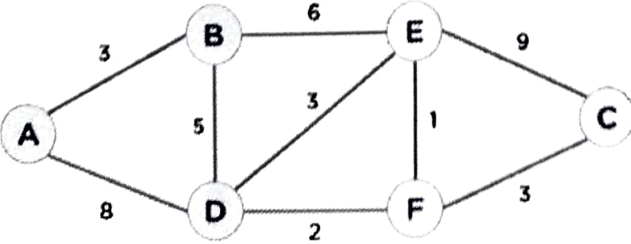
TERM : 3 rd March 2025-21 st June 2025		COURSE NAME: Design and Analysis of Algorithms
DATE : 17-06-2025	TIME:9:15am-10:15am	COURSE CODE : IS43
MAX MARKS: 30	Semester: 4	PORTIONS : L 17 – L42
Faculty Signature with initials (All faculty handling the subject)		

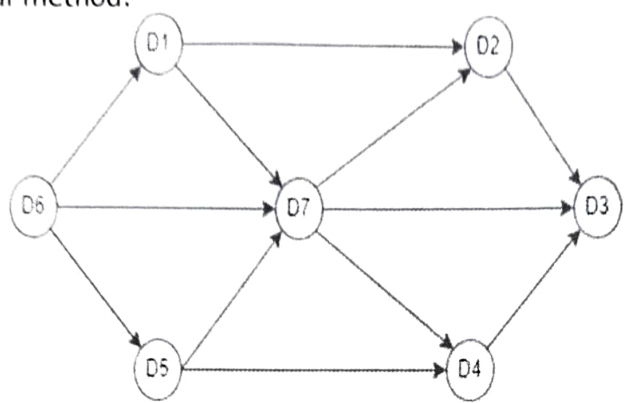
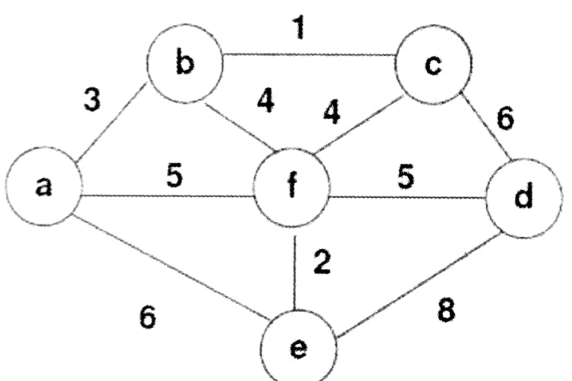


Mobile Phones are banned

Instructions to Candidates: **Answer any TWO full questions.**

Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1 a.	Apply Warshall's algorithm to find transitive closure for the below given graph. 	L3	CO3	5
b	Give the comparison between BFS and DFS with an example problem.	L2	CO4	5
c	Construct state space tree for solving the four Queen's problem by back tracking.	L3	CO5	5
2 a.	Apply Kruskal's algorithm to construct a minimum spanning tree for the following graph. 	L3	CO3	5

2 b.	Obtain topological sorting for a given graph by using source removal method. 	L3	CO4	5										
2 c.	Apply Heapsort for the list [9,7,1,8,3,6,2,4,10,5] using Bottom up approach.	L3	CO5	5										
3 a.	Apply Dijkstras algorithm to find the single source shortest path problem with vertex "a" as the source. 	L4	CO3	5										
b.	A message consisting of the characters given in the table below has to be transmitted over a network in secure manner. <table border="1" data-bbox="301 1428 819 1512"> <tr> <td>Character</td><td>A</td><td>M</td><td>R</td><td>_</td></tr> <tr> <td>Probability</td><td>0.4</td><td>0.2</td><td>0.3</td><td>0.1</td></tr> </table> i. Encode the text RAMA_RAM using Huffman codes ii. Decode the text whose encoding is 1000101	Character	A	M	R	_	Probability	0.4	0.2	0.3	0.1	L3	CO4	5
Character	A	M	R	_										
Probability	0.4	0.2	0.3	0.1										
c.	Define the following i. P problem ii. NP problem iii. NP complete problem	L2	CO5	5										

*L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create